

# Connected High-Event Registry

Analytic ledger replacement, Packet C

## Abstract

This note replaces the exact-executable arithmetic used in the high spherical-shell staircase by displayed rational witnesses. It contains the complete seventeen-face localization registry, the three algebraic splits, the three dashed (impossible) cap cells, the five global checkpoint payments, the seven conditional payments, and the corrected cap-74 path. It also contains the order-free closed-post-event theorem which maps every high-staircase state to a payment at its own event wall. That theorem, rather than the historical executable row census, is the finite-combinatorial closure used by the proof. All comparisons are strict rational comparisons after the two elementary bounds  $333/106 < \pi < 22/7$ . In particular, no decimal approximation, interval package, or external ledger is a logical input to this packet.

## 1 Definitions and strict proxy convention

For  $0 < \rho < 1$ , put

$$z = z_\rho := \frac{\pi}{1-\rho}, \quad x = x_\rho := z_\rho^2, \quad k_m(\rho) := \sqrt{x_\rho + m(m+1)}. \quad (1)$$

The three-dimensional Weyl term is

$$\mathcal{W}(\rho, K) := \frac{2}{9\pi}(1-\rho^3)K^3. \quad (2)$$

At  $K = k_m(\rho)$ , the product part of the strict channel proxy for the mode  $(\ell, n)$ ,  $n \geq 2$ , is

$$n^2x + \ell(\ell+1) < x + m(m+1).$$

It is therefore governed by

$$\Delta_{m;n,\ell}(\rho) := m(m+1) - \ell(\ell+1) - (n^2-1)x_\rho. \quad (3)$$

The channel is product-possible only when  $\Delta_{m;n,\ell} > 0$ . Equality is excluded because the spectral proxy is strict. Since

$$x'_\rho = \frac{2\pi^2}{(1-\rho)^3} > 0, \quad \Delta'_{m;n,\ell} = -(n^2-1)x'_\rho < 0, \quad (4)$$

a positive witness at  $r$  controls the side  $\rho \leq r$ , while a negative witness controls the side  $\rho \geq r$ .

We write  $H$  for the largest first-mode angular index retained in a cap cell and  $h$  for the largest second-mode angular index. A complete block through  $H$  has multiplicity  $(H+1)^2$ , because  $\sum_{\ell=0}^H (2\ell+1) = (H+1)^2$ .

## 2 The only transcendental enclosure

**Lemma 1** (Rational enclosure of  $\pi$ ). *One has*

$$\pi_- := \frac{333}{106} < \pi < \frac{22}{7} =: \pi_+. \quad (5)$$

*Proof.* For  $0 < u < 1$ , the alternating expansion of  $\arctan u$  lies between each pair of consecutive partial sums. The tangent addition formula gives

$$\tan\left(4 \arctan \frac{1}{5} - \arctan \frac{1}{239}\right) = 1;$$

indeed,  $\tan(4 \arctan(1/5)) = 120/119$ , so both the numerator and denominator in the last tangent quotient are  $28561/(119 \cdot 239)$ . The angle lies in  $(0, \pi/2)$ , and hence Machin's identity follows:

$$\pi = 16 \arctan \frac{1}{5} - 4 \arctan \frac{1}{239}.$$

Using an even partial sum for the positive term and an upper partial sum for the subtracted term gives

$$\begin{aligned} \pi &> 16 \left( \frac{1}{5} - \frac{1}{3 \cdot 5^3} + \frac{1}{5 \cdot 5^5} - \frac{1}{7 \cdot 5^7} \right) - \frac{4}{239}, \\ [\text{right side}] - \frac{333}{106} &= \frac{3418213}{41563593750} > 0. \end{aligned}$$

Likewise,

$$\begin{aligned} \pi &< 16 \left( \frac{1}{5} - \frac{1}{3 \cdot 5^3} + \frac{1}{5 \cdot 5^5} \right) - 4 \left( \frac{1}{239} - \frac{1}{3 \cdot 239^3} \right), \\ \frac{22}{7} - [\text{right side}] &= \frac{1845738322}{1493178640625} > 0. \end{aligned}$$

This proves (5). □

## 3 The seventeen rational localization faces

Let

$$\rho_c := \frac{1}{1 + 2\pi}.$$

Because  $\pi > 3$ ,  $\rho_c < 1/7$ . The seventeen registry ratios are strictly ordered as follows:

$$\begin{aligned} \frac{4}{25} &< \frac{1}{5} < \frac{1}{4} < \frac{3}{10} < \frac{1}{3} < \frac{7}{20} < \frac{3}{8} < \frac{2}{5} < \frac{5}{12} < \frac{3}{7} < \frac{1}{2} \\ &< \frac{8}{15} < \frac{107}{200} < \frac{3}{5} < \frac{16}{25} < \frac{13}{20} < \frac{2}{3}. \end{aligned} \quad (6)$$

For a completely explicit distinctness check, the sixteen consecutive gaps in (6) are

$$\frac{1}{25}, \frac{1}{20}, \frac{1}{20}, \frac{1}{30}, \frac{1}{60}, \frac{1}{40}, \frac{1}{40}, \frac{1}{60}, \frac{1}{84}, \frac{1}{14}, \frac{1}{30}, \frac{1}{600}, \frac{13}{200}, \frac{1}{25}, \frac{1}{100}, \frac{1}{60}. \quad (7)$$

Moreover

$$\frac{4}{25} - \frac{1}{7} = \frac{3}{175} > 0, \quad \frac{7}{8} - \frac{2}{3} = \frac{5}{24} > 0.$$

Thus all seventeen ratios lie in the high strip  $\rho_c < \rho < 7/8$ .

For a positive row below, we use

$$\Delta_{m;n,\ell}(r) > m(m+1) - \ell(\ell+1) - (n^2 - 1) \left( \frac{\pi_+}{1-r} \right)^2.$$

For a negative row, we use

$$-\Delta_{m;n,\ell}(r) > (n^2 - 1) \left( \frac{\pi_-}{1-r} \right)^2 - m(m+1) + \ell(\ell+1).$$

The resulting exact witnesses are the following. “Retain” means only that the product wall does not permit deletion of the indicated block; the independent Bessel wall may still remove it.

| no. | $r$     | $(m; n, \ell)$ | exact sign witness           | one-sided cap/event consequence                               |
|-----|---------|----------------|------------------------------|---------------------------------------------------------------|
| 1   | 1/5     | (11; 3, 1)     | $\Delta > 320/49$            | retain the third block 0:1 for $\rho \leq 1/5$                |
| 2   | 3/10    | (8; 2, 3)      | $-\Delta > 58215/137641$     | at $k_8$ , $h \leq 2$ for $\rho \geq 3/10$                    |
| 3   | 1/3     | (8; 2, 2)      | $-\Delta > 27699/44944$      | at $k_8$ , $h \leq 1$ for $\rho \geq 1/3$                     |
| 4   | 3/8     | (9; 2, 3)      | $\Delta > 2622/1225$         | retain the $h = 3$ column below this face                     |
| 5   | 2/5     | (9; 2, 2)      | $\Delta > 248/147$           | retain the $h = 2$ column below this face                     |
| 6   | 5/12    | (9; 2, 1)      | $\Delta > 2200/2401$         | retain the $h = 1$ column below this face                     |
| 7   | 3/7     | (9; 2, 0)      | $-\Delta > 120843/179776$    | no second mode at $k_9$ on the upper side                     |
| 8   | 1/2     | (10; 2, 0)     | $-\Delta > 23677/2809$       | no second mode at $k_{10}$ on the upper side                  |
| 9   | 107/200 | (11; 2, 0)     | $-\Delta > 13302732/2699449$ | no second mode at $k_{11}$ on the upper side                  |
| 10  | 8/15    | (11; 2, 0)     | $-\Delta > 2175627/550564$   | the clean stated cutoff: a second mode forces $\rho < 8/15$   |
| 11  | 3/5     | (8; 2, 0)      | $-\Delta > 5080707/44944$    | select the no-second $k_8$ cap/payment side                   |
| 12  | 16/25   | (9; 2, 0)      | $-\Delta > 1555635/11236$    | select the no-second $k_9$ cap/payment side                   |
| 13  | 13/20   | (10; 2, 0)     | $-\Delta > 18126190/137641$  | select the no-second $k_{10}$ cap/payment side                |
| 14  | 2/3     | (11; 2, 0)     | $-\Delta > 1510851/11236$    | select the no-second $k_{11}$ cap/payment side                |
| 15  | 4/25    | (10; 3, 0)     | $-\Delta > 260740/137641$    | no third mode at $k_{10}$ on the upper side                   |
| 16  | 1/4     | (11; 3, 0)     | $-\Delta > 23484/2809$       | a third mode at $k_{11}$ forces $\rho < 1/4$                  |
| 17  | 7/20    | (9; 2, 4)      | $-\Delta > 36230/474721$     | the $H = 6, h = 4$ cap-74 cell can occur only below this face |

Every entry is a displayed positive rational number. Together with the monotonicity (4), the table proves the asserted side on an interval, not merely at a sampled point.

## 4 Three algebraic splits and the five $k_{11}$ implications

At  $\rho = \rho_c$ ,  $x = (\pi + 1/2)^2$ , while at  $\rho = 7/8$ ,  $x = 64\pi^2$ . Lemma 1 gives the exact chain

$$(\pi + \tfrac{1}{2})^2 < \left(\frac{51}{14}\right)^2 = \frac{2601}{196} < 16 < \frac{68}{3} < 34 < 64 \cdot 3^2 < 64\pi^2. \quad (8)$$

Consequently each of  $x = 16, 68/3, 34$  is attained at a unique point in the high strip. The relevant channel arithmetic is

| $x$  | channel                     | exact value of $\Delta$          |
|------|-----------------------------|----------------------------------|
| 16   | $(m; n, \ell) = (11; 3, 1)$ | $132 - 2 - 8 \cdot 16 = 2 > 0$ , |
| 68/3 | $(10; 2, 6)$                | $110 - 42 - 3(68/3) = 0$ ,       |
| 34   | $(11; 2, 5)$                | $132 - 30 - 3 \cdot 34 = 0$ .    |

(9)

Thus the first split lies on the possible side. The latter two are exact channel walls, and their new modes are excluded by the strict convention.

The two localization consequences at  $k_{11}$  may also be read without the table:

$$x_{8/15} - 44 > \left(\frac{\pi_-}{1 - 8/15}\right)^2 - 44 = \frac{725209}{550564} > 0, \quad (10)$$

$$x_{1/4} - \frac{33}{2} > \left(\frac{\pi_-}{1 - 1/4}\right)^2 - \frac{33}{2} = \frac{5871}{5618} > 0. \quad (11)$$

Hence a second mode at  $k_{11}$  forces  $\rho < 8/15$ , and a third mode forces  $\rho < 1/4$ . Finally, the first-mode ball walls give

$$\left(\frac{69}{5}\right)^2 - 132 = \frac{1461}{25} = 44 + \frac{361}{25} = \frac{33}{2} + \frac{2097}{50}, \quad (12)$$

$$\left(\frac{64}{5}\right)^2 - 132 = \frac{796}{25} = \frac{33}{2} + \frac{767}{50}. \quad (13)$$

If  $H = 10$  is present at  $k_{11}$ , then  $x$  is larger than both the second threshold 44 and the third threshold  $33/2$ ; hence every second and third mode is excluded. If  $H = 9$  is present, every third mode is excluded. Equations (10)–(13) are the five formerly separate  $k_{11}$  localization implications.

## 5 The three impossible cap cells

If the first mode  $H$  is counted at  $k_m$ , its strict ball wall  $\beta_{H,1} < k_m$  forces  $x > \beta_{H,1}^2 - m(m+1)$ . If the second mode  $h$  is counted, its product wall forces  $x < (m(m+1) - h(h+1))/3$ . The three dashed cells are therefore empty by the following exact, incompatible bounds:

| cell                    | required lower bound      | required upper bound | gap      |
|-------------------------|---------------------------|----------------------|----------|
| $k_8 : H = 6, h \geq 2$ | $100 - 72 = 28$           | $(72 - 6)/3 = 22$    | 6        |
| $k_9 : H = 8, h \geq 0$ | $(71/6)^2 - 90 = 1801/36$ | $90/3 = 30$          | $721/36$ |
| $k_9 : H = 7, h \geq 3$ | $(23/2)^2 - 90 = 169/4$   | $(90 - 12)/3 = 26$   | $65/4$ . |

(14)

No limiting equality can rescue a cell: both necessary modal inequalities are strict.

## 6 Checkpoint monotonicity and five global payments

Put  $M = m(m+1)$  and  $\mathcal{W}_m(\rho) := \mathcal{W}(\rho, k_m(\rho))$ . Logarithmic differentiation shows that  $\mathcal{W}'_m$  has the sign of

$$\pi^2(1+\rho) - M\rho^2(1-\rho)^2. \quad (15)$$

For  $m \leq 11$ ,

$$\pi^2(1+\rho) - M\rho^2(1-\rho)^2 > 9 - \frac{132}{16} = \frac{3}{4} > 0.$$

Thus every  $\mathcal{W}_m$ ,  $m \leq 11$ , is strictly increasing. This single argument replaces both executable derivative-bound predicates.

At  $\rho_c$ , one has  $z = \pi + 1/2 > 193/53$ ,  $\rho_c < 1/7$ , and  $1/\pi > 7/22$ . Consequently

$$\mathcal{W}(\rho_c, K) > \frac{2}{9} \frac{7}{22} \left(1 - \frac{1}{7^3}\right) K^3 = \frac{38}{539} K^3. \quad (16)$$

For each row in the next table, the middle column proves  $k_m(\rho_c) > t_m$ , and the last column proves  $\mathcal{W}_m(\rho_c) > C_m$ .

| $m$ | cap $C_m$ | $t_m$   | $(193/53)^2 + M - t_m^2$ | $(38/539)t_m^3 - C_m$ |
|-----|-----------|---------|--------------------------|-----------------------|
| 7   | 40        | 208/25  | 67049/1755625            | 5083656/8421875       |
| 8   | 53        | 923/100 | 1901439/28090000         | 656778873/269500000   |
| 9   | 73        | 254/25  | 61431/1755625            | 7911557/8421875       |
| 10  | 89        | 111/10  | 14211/280900             | 1999489/269500        |
| 11  | 121       | 241/20  | 65271/1123600            | 5076899/2156000       |

Every last-column entry is positive. Monotonicity therefore extends each global payment from  $\rho_c$  to the complete high strip.

## 7 Seven conditional first-block payments

We now give a general reduction which avoids any numerical enclosure of a radical. Suppose the presence of first mode  $H$  below the checkpoint  $k_m$  forces  $K > t$ . Set

$$a^2 := t^2 - m(m+1), \quad \rho_0 := 1 - \frac{\pi}{a}. \quad (17)$$

All seven rows below have  $a > \pi$ : the smallest listed radicand is

$$\frac{89}{4}, \quad \frac{89}{4} - \left(\frac{22}{7}\right)^2 = \frac{2425}{196} > 0.$$

If the mode is present for some  $K \leq k_m(\rho)$ , then  $k_m(\rho) > t$ , hence  $x_\rho > a^2$  and  $\rho > \rho_0$ . By the strict increase of  $\mathcal{W}_m$ ,

$$\mathcal{W}_m(\rho) > \mathcal{W}_m(\rho_0) = \mathcal{W}(\rho_0, t).$$

Choose a rational  $r > \rho_0$ . Since  $\mathcal{W}(\rho, t)$  decreases with  $\rho$  at fixed  $t$ , Lemma 1 yields

$$\mathcal{W}(\rho_0, t) > \mathcal{W}(r, t) > \frac{7}{99}(1-r^3)t^3. \quad (18)$$

The condition  $r > \rho_0$  follows, without taking a square root, from

$$L := \pi_-^2 - (1-r)^2 a^2 > 0. \quad (19)$$

Here is the complete calculation. The column  $P$  is the reserve in (18) above the stated cap.

| $m$ | $H$ | $t$  | cap | $a^2$   | $r$  | $L$             | $P = \frac{7}{99}(1-r^3)t^3 - \text{cap}$ |
|-----|-----|------|-----|---------|------|-----------------|-------------------------------------------|
| 7   | 6   | 10   | 53  | 44      | 8/15 | 725209/2528100  | 18659/2673                                |
| 8   | 7   | 23/2 | 80  | 241/4   | 3/5  | 64349/280900    | 213281/49500                              |
| 9   | 8   | 71/6 | 81  | 1801/36 | 3/5  | 4713989/2528100 | 14506973/1336500                          |
| 10  | 7   | 23/2 | 100 | 89/4    | 7/20 | 2105431/4494400 | 18539033/6336000                          |
| 10  | 8   | 71/6 | 97  | 1081/36 | 3/7  | 317585/4955076  | 2865431/261954                            |
| 10  | 9   | 64/5 | 100 | 1346/25 | 3/5  | 8811001/7022500 | 25143284/1546875                          |
| 11  | 9   | 64/5 | 125 | 796/25  | 1/2  | 536261/280900   | 58757/12375                               |

All  $L$ 's and  $P$ 's are positive. Each row therefore proves both the localization of its conditional branch and its Weyl payment.

## 8 The connected cap-74 path

The seventeenth localization row is tied to the exceptional  $k_9, H = 6, h = 4$  cap as follows. Angular propagation from the strict (3, 2) Bessel wall gives

$$j_{9/2,2}^2 > \left(\frac{103}{10}\right)^2 + 20 - 12 = \frac{11409}{100} = 108 + \frac{609}{100}. \quad (20)$$

Accordingly the strict proxy uses  $\beta_{4,2}^2 = 11409/100$ ; the strict inequality belongs to the actual Bessel zero  $j_{9/2,2}$ , not to the proxy constant. At  $k_9$ , the product wall for (4, 2) is

$$3x < 90 - 20 = 70, \quad \text{i.e.} \quad x < \frac{70}{3}.$$

The ball wall (20), on the other hand, forces

$$x > \frac{11409}{100} - 90 = \frac{2409}{100}, \quad \frac{2409}{100} - \frac{70}{3} = \frac{227}{300} > 0.$$

Thus the fully sharpened cap-74 cell is empty.

For completeness, even the deliberately conservative cell is paid. At  $\rho = 7/20$ ,

$$x_{7/20} - \frac{70}{3} > \frac{400}{169} \left(\frac{333}{106}\right)^2 - \frac{70}{3} = \frac{36230}{1424163} > 0, \quad (21)$$

so a counted (4, 2) mode forces  $\rho < 7/20$ . Also

$$108 - \left(\frac{207}{20}\right)^2 = \frac{351}{400} > 0, \quad (22)$$

so its ball wall forces  $K > 207/20$ . Therefore

$$\begin{aligned} \mathcal{W}(\rho, K) &> \frac{7}{99} \left(1 - \left(\frac{7}{20}\right)^3\right) \left(\frac{207}{20}\right)^3 \\ &= \frac{52823261673}{704000000} = 74 + \frac{727261673}{704000000} > 74. \end{aligned} \quad (23)$$

The cap itself is exactly  $(6+1)^2 + (4+1)^2 = 49 + 25 = 74$ . Equations (20)–(23) cover both the actual incompatibility and every row of the conservative cap-74 fallback.

## 9 Order-free exhaustion and payment of every high event

This section supplies the finite-combinatorial implication which is not contained in a list of checkpoint payments alone. It is deliberately independent of the order in which two modal walls cross.

For  $7 \leq m \leq 11$ , set

$$\mathfrak{B}_m^+ := \{(\rho, K) : \rho_c \leq \rho \leq 39/50, k_{m-1}(\rho) \leq K < k_m(\rho)\}. \quad (24)$$

The final spectral band is closed on the right:

$$\mathfrak{B}_{11}^{\text{sp}} := \{(\rho, K) : \rho_c \leq \rho \leq 39/50, k_{10}(\rho) \leq K \leq k_{11}(\rho)\}.$$

At  $K = k_{11}$  we use the original strict proxy. On the half-open event bands we use

$$b_{\ell,n}(z)^2 = \max\{n^2 z^2 + \ell(\ell+1), \beta_{\ell,n}^2\}, \quad C_z(K) = \sum_{\ell,n} (2\ell+1) \mathbf{1}_{\{b_{\ell,n}(z) < K\}},$$

and define the closed post-event count

$$\hat{C}_z(K) := \sum_{(\ell,n)} (2\ell+1) \mathbf{1}_{\{b_{\ell,n}(z) \leq K\}}. \quad (25)$$

The sum is over the checkpoint inventory appropriate to the upper endpoint of the band. Thus a simultaneous event is charged with all of its entering multiplicities before it is paid.

For a post-event state let  $H$  be the largest first-channel angular index, let  $h = -1$  if no second channel is present and otherwise let  $h$  be its largest angular index, and put  $\tau = 1$  if a third channel is present and  $\tau = 0$  otherwise. Missing angular indices only lower the count, since

$$\hat{C}_z(K) \leq (H+1)^2 + \mathbf{1}_{\{h \geq 0\}}(h+1)^2 + 4\tau. \quad (26)$$

The exhaustive checkpoint inventories are

| $m$ | $H$         | $h$        | third channels |
|-----|-------------|------------|----------------|
| 7   | $H \leq 6$  | $h \leq 1$ | none           |
| 8   | $H \leq 7$  | $h \leq 3$ | none           |
| 9   | $H \leq 8$  | $h \leq 4$ | none           |
| 10  | $H \leq 9$  | $h \leq 5$ | $\ell = 0, 1$  |
| 11  | $H \leq 10$ | $h \leq 4$ | $\ell = 0, 1.$ |

(27)

First channels with  $\ell \geq m$  are excluded at  $k_m$  by their product wall. For the first omitted second channel, the two necessary strict bounds are

| $m$                  | 7          | 8              | 9              | 10             | 11              |
|----------------------|------------|----------------|----------------|----------------|-----------------|
| first omitted $\ell$ | 2          | 4              | 5              | 6              | 5               |
| product wall         | $x < 50/3$ | $x < 52/3$     | $x < 20$       | $x < 68/3$     | $x < 34$        |
| ball wall            | $x > 25$   | $x > 4209/100$ | $x > 7641/100$ | $x > 6841/100$ | $x > 3441/100.$ |

The  $m = 10$  ball wall follows from angular propagation of the  $\ell = 5, n = 2$  registry wall. In every column the bounds are incompatible; for larger  $\ell$ , the product upper bound decreases and the propagated ball lower bound increases.

For  $n = 3$ , the product wall gives  $x \leq 45/4 < (193/53)^2 < x$  through  $m = 9$ , and at  $m = 10$  every  $\ell \geq 2$  gives  $x \leq 13 < (193/53)^2$ . At  $m = 11$ , the first omitted third channel  $\ell = 2$  would require

$$x < 63/4, \quad x > (61/5)^2 - 132 = 421/25.$$

Angular propagation removes the rest of that tail. For  $n \geq 4$ , the least restrictive omitted case is  $m = 11, \ell = 0, n = 4$ , which would require  $x < 44/5 < (193/53)^2$ . This proves (27) without an unprinted search.

The strict simultaneous restrictions used below are

| $m$ | restriction                         |                                      |
|-----|-------------------------------------|--------------------------------------|
| 8   | $H = 6 \implies h \leq 1,$          | $H = 7$ unrestricted by this row,    |
| 9   | $H = 8 \implies h = -1,$            | $H = 7 \implies h \leq 2,$           |
| 10  | $H = 9 \implies h = -1, \tau = 0,$  | $H = 8 \implies h \leq 3, \tau = 0,$ |
| 11  | $H = 10 \implies h = -1, \tau = 0,$ | $H = 9 \implies \tau = 0.$           |

(28)

For  $m = 8, 9$  these are exactly the incompatible cells in (14). For  $m = 10$ , the three first-wall lower bounds are  $1346/25, 1081/36, 89/4$ , which respectively exceed the applicable second or third thresholds  $110/3, 30, 55/4$ . For  $m = 11$ , (12)–(13) give the last two restrictions. All gaps are strict, so the same restrictions hold for the closed post-event count on the half-open bands.

### 9.1 The fourteen localization witnesses

If a closed post-event second mode  $h$  is present below  $k_m$ , then

$$4x + h(h+1) \leq K^2 < x + m(m+1), \quad 3x < m(m+1) - h(h+1).$$

At the strict spectral upper face the first inequality is strict and the same conclusion follows. Likewise, a third mode forces  $8x < m(m+1)$ . The following table contains every localization used in the state table. Its last column is  $(333/(106(1-r)))^2 - A > 0$ , where  $x < A$  is the forced bound.

| band                  | state implication   | $A$   | $r$  | exact positive witness |
|-----------------------|---------------------|-------|------|------------------------|
| $\mathfrak{B}_7^+$    | some second channel | 56/3  | 3/10 | 608779/412923          |
| $\mathfrak{B}_8^+$    | some second channel | 24    | 3/8  | 88824/70225            |
| $\mathfrak{B}_8^+$    | $h \geq 2$          | 22    | 1/3  | 9233/44944             |
| $\mathfrak{B}_8^+$    | $h = 3$             | 20    | 3/10 | 19405/137641           |
| $\mathfrak{B}_9^+$    | some second channel | 30    | 3/7  | 40281/179776           |
| $\mathfrak{B}_9^+$    | $h \geq 2$          | 28    | 5/12 | 138056/137641          |
| $\mathfrak{B}_9^+$    | $h \geq 3$          | 26    | 2/5  | 15889/11236            |
| $\mathfrak{B}_9^+$    | $h = 4$             | 70/3  | 7/20 | 36230/1424163          |
| $\mathfrak{B}_{10}^+$ | some second channel | 110/3 | 1/2  | 23677/8427             |
| $\mathfrak{B}_{10}^+$ | $h \geq 4$          | 30    | 3/7  | 40281/179776           |
| $\mathfrak{B}_{10}^+$ | $h = 5$             | 80/3  | 2/5  | 25195/33708            |
| $\mathfrak{B}_{10}^+$ | $\tau = 1$          | 55/4  | 4/25 | 65185/275282           |
| $\mathfrak{B}_{11}^+$ | some second channel | 44    | 8/15 | 725209/550564          |
| $\mathfrak{B}_{11}^+$ | $\tau = 1$          | 33/2  | 1/4  | 5871/5618              |

Every defining ratio face is therefore on the excluded side of its state; the directions in this table are interval implications, not sampled decisions.

### 9.2 Three payment rules

Put  $F_j(\rho) := \mathcal{W}(\rho, k_j(\rho))$ , and define

$$P(C; c, r) := \frac{7}{99}(1-r^3)c^3 - C, \quad Q_j(c, r) := \left( \frac{333}{106(1-r)} \right)^2 + j(j+1) - c^2. \quad (29)$$



The derivative calculation in (15) gives, uniformly for  $6 \leq j \leq 10$ ,

$$F'_j(\rho) > 0, \quad 9 - \frac{110}{16} = \frac{17}{8} > 0,$$

and  $\mathcal{W}(\rho, c)$  decreases with  $\rho$  at fixed  $c > 0$ . Together with  $2/(9\pi) > 7/99$ , this gives three rules.

*L.* If a state forces  $K \geq c$  and  $\rho < r$ , then  $P(C; c, r) > 0$  implies  $\mathcal{W}(\rho, K) > C$ .

*B.* If a state forces  $K \geq c$  in  $\mathfrak{B}_m^+$ , and  $P(C; c, r) > 0$ ,  $Q_{m-1}(c, r) > 0$ , then it is paid on both sides of  $r$ . For  $\rho \leq r$ , use the fixed wall  $c$ . For  $\rho \geq r$ , use  $K \geq k_{m-1}(\rho)$ , the increase of  $F_{m-1}$ , and  $k_{m-1}(r) > c$ . In particular,  $F_{m-1}(r) = \mathcal{W}(r, k_{m-1}(r)) > \mathcal{W}(r, c) > C$ . Both arguments are strict at  $\rho = r$ .

*G.* If the state cap is at most a displayed global cap  $C$ , then  $K \geq k_{m-1}(\rho)$  and the strict global lower-checkpoint payment pays it, including the lower checkpoint itself.

There is one initial checkpoint anchor of type  $B_0$ . Since  $\pi < 7/2$ , one has  $\rho_c > 1/8$ . The increase of  $F_6$ , together with the  $B_{7,0}$  row below, gives

$$F_6(\rho) > F_6(1/8) > \mathcal{W}(1/8, 73/10) > 26.$$

### 9.3 The disjoint thirty-eight-state map

The following rows are literally disjoint. In a  $G$ -row the displayed state is the exact complement of the exceptional rows in its band, not a second overlapping estimate. The cap column is allowed to dominate more than one exact block sum, but every exact post-event cap is obtained from (26) and is one of the block sums represented here.

| row        | band | disjoint post-event state                                                                         | cap $C$ | $c$    | $r$  |
|------------|------|---------------------------------------------------------------------------------------------------|---------|--------|------|
| $B_{7,0}$  | 7    | $H \leq 4, -1 \leq h \leq 0$                                                                      | 26      | 73/10  | 1/8  |
| $L_{7,1}$  | 7    | $H \leq 4, h = 1$                                                                                 | 29      | 77/10  | 3/10 |
| $B_{7,1}$  | 7    | $H = 5, h = -1$                                                                                   | 36      | 87/10  | 1/2  |
| $L_{7,2}$  | 7    | $H = 5, 0 \leq h \leq 1$                                                                          | 40      | 87/10  | 3/10 |
| $B_{7,2}$  | 7    | $H = 6, h = -1$                                                                                   | 49      | 10     | 3/5  |
| $L_{7,3}$  | 7    | $H = 6, 0 \leq h \leq 1$                                                                          | 53      | 10     | 3/10 |
| $G_8$      | 8    | $H \leq 4, h \leq 2$ , or $H = 5, h \leq 1$                                                       | 40      | —      | —    |
| $L_{8,1}$  | 8    | $H = 5, h = 2$                                                                                    | 45      | 9      | 1/3  |
| $L_{8,2}$  | 8    | $H \leq 5, h = 3$                                                                                 | 52      | 103/10 | 3/10 |
| $B_{8,1}$  | 8    | $H = 6, h = -1$                                                                                   | 49      | 10     | 3/5  |
| $L_{8,3}$  | 8    | $H = 6, 0 \leq h \leq 1$                                                                          | 53      | 10     | 3/8  |
| $B_{8,2}$  | 8    | $H = 7, h = -1$                                                                                   | 64      | 23/2   | 2/3  |
| $L_{8,4}$  | 8    | $H = 7, 0 \leq h \leq 3$                                                                          | 80      | 23/2   | 3/8  |
| $G_9$      | 9    | $H \leq 4$ ; or $H = 5, h \leq 3$ ; or $H = 6, h \leq 1$                                          | 53      | —      | —    |
| $L_{9,1}$  | 9    | $H = 6, h = 2$                                                                                    | 58      | 10     | 5/12 |
| $L_{9,2}$  | 9    | $H = 6, h = 3$                                                                                    | 65      | 103/10 | 2/5  |
| $L_{9,3}$  | 9    | $5 \leq H \leq 6, h = 4$                                                                          | 74      | 207/20 | 7/20 |
| $B_{9,1}$  | 9    | $H = 7, h = -1$                                                                                   | 64      | 23/2   | 2/3  |
| $L_{9,4}$  | 9    | $H = 7, 0 \leq h \leq 2$                                                                          | 73      | 23/2   | 3/7  |
| $B_{9,2}$  | 9    | $H = 8, h = -1$                                                                                   | 81      | 71/6   | 2/3  |
| $G_{10}$   | 10   | $H \leq 4$ ; or $H = 5$ except $(h, \tau) = (5, 1)$ ; or $H = 6, h \leq 3$ ; or $H = 7, h \leq 2$ | 73      | —      | —    |
| $L_{10,1}$ | 10   | $H = 6, h = 4, \tau = 0$                                                                          | 74      | 53/5   | 3/7  |
| $L_{10,2}$ | 10   | $H = 6, h = 4, \tau = 1$                                                                          | 78      | 53/5   | 4/25 |
| $L_{10,3}$ | 10   | $H = 6, h = 5, \tau = 0$                                                                          | 85      | 11     | 2/5  |
| $L_{10,4}$ | 10   | $5 \leq H \leq 6, h = 5, \tau = 1$                                                                | 89      | 11     | 4/25 |

| row        | band | disjoint post-event state                                                  | cap $C$ | $c$  | $r$   |
|------------|------|----------------------------------------------------------------------------|---------|------|-------|
| $L_{10,5}$ | 10   | $H = 7, h = 3$                                                             | 80      | 23/2 | 1/2   |
| $L_{10,6}$ | 10   | $H = 7, h = 4$                                                             | 89      | 23/2 | 3/7   |
| $L_{10,7}$ | 10   | $H = 7, h = 5$                                                             | 100     | 23/2 | 2/5   |
| $B_{10,1}$ | 10   | $H = 8, h = -1$                                                            | 81      | 71/6 | 3/5   |
| $L_{10,8}$ | 10   | $H = 8, 0 \leq h \leq 3$                                                   | 97      | 71/6 | 1/2   |
| $B_{10,2}$ | 10   | $H = 9, h = -1$                                                            | 100     | 64/5 | 16/25 |
| $G_{11}$   | 11   | $H \leq 6$ ; or $H = 7$ except $(h, \tau) = (4, 1)$ ; or $H = 8, h \leq 1$ | 89      | —    | —     |
| $L_{11,1}$ | 11   | $H = 7, h = 4, \tau = 1$                                                   | 93      | 23/2 | 1/4   |
| $B_{11,1}$ | 11   | $H = 8, 2 \leq h \leq 4, \tau = 0$                                         | 106     | 71/6 | 3/7   |
| $L_{11,2}$ | 11   | $H = 8, 2 \leq h \leq 4, \tau = 1$                                         | 110     | 71/6 | 1/4   |
| $B_{11,2}$ | 11   | $H = 9, h = -1$                                                            | 100     | 64/5 | 3/5   |
| $L_{11,3}$ | 11   | $H = 9, 0 \leq h \leq 4$                                                   | 125     | 64/5 | 8/15  |
| $B_{11,3}$ | 11   | $H = 10, h = -1$                                                           | 121     | 69/5 | 13/20 |

Apart from the checkpoint anchor  $B_{7,0}$ , the wall  $c$  in each exceptional row is forced by the corresponding fixed part of  $b_{\ell,n}$ . Besides the displayed first and second walls, the only weakenings are

$$\beta_{4,2}^2 - (207/20)^2 = 2787/400 > 0, \quad \beta_{4,2}^2 - (53/5)^2 = 173/100 > 0, \quad \beta_{5,2} - 11 = 19/10 > 0.$$

#### 9.4 The twenty-two localized payments

For each  $L$ -row, the appropriate entry of the fourteen-row table gives  $\rho < r$ , and the fixed wall gives  $K \geq c$ . Direct substitution in (29) gives the following complete list.

| row        | $P(C; c, r)$        | row        | $P(C; c, r)$        |
|------------|---------------------|------------|---------------------|
| $L_{7,1}$  | 21676933/9000000    | $L_{7,2}$  | 58340437/11000000   |
| $L_{7,3}$  | 1564/99             | $L_{8,1}$  | 51/11               |
| $L_{8,2}$  | 2294563597/99000000 | $L_{8,3}$  | 88567/6336          |
| $L_{8,4}$  | 8866645/405504      | $L_{9,1}$  | 162353/21384        |
| $L_{9,2}$  | 10063157/1375000    | $L_{9,3}$  | 727261673/704000000 |
| $L_{9,4}$  | 252947/9702         | $L_{10,1}$ | 2173382/606375      |
| $L_{10,2}$ | 126077081/21484375  | $L_{10,3}$ | 386/125             |
| $L_{10,4}$ | 73838/15625         | $L_{10,5}$ | 89303/6336          |
| $L_{10,6}$ | 97715/9702          | $L_{10,7}$ | 7197/11000          |
| $L_{10,8}$ | 943655/171072       | $L_{11,1}$ | 72407/5632          |
| $L_{11,2}$ | 810599/152064       | $L_{11,3}$ | 32924779/41765625   |

All twenty-two fractions are positive, so rule  $L$  pays the dominant post-event cap in every localized row.

#### 9.5 The twelve bridge pairs and four global rows

The eleven ordinary  $B$ -rows and the initial anchor  $B_{7,0}$  have the following exact  $P, Q$  pairs. The subscript on  $Q$  is the lower checkpoint index.

| row       | $P(C; c, r)$      | $Q_j(c, r)$               |
|-----------|-------------------|---------------------------|
| $B_{7,0}$ | 73625809/50688000 | $Q_6 = 22025711/13764100$ |
| $B_{7,1}$ | 417183/88000      | $Q_6 = 1625379/280900$    |
| $B_{7,2}$ | 637/99            | $Q_6 = 165473/44944$      |
| $B_{8,1}$ | 637/99            | $Q_7 = 794689/44944$      |

| row        | $P(C; c, r)$       | $Q_j(c, r)$                |
|------------|--------------------|----------------------------|
| $B_{8,2}$  | 249635/21384       | $Q_7 = 35314/2809$         |
| $B_{9,1}$  | 249635/21384       | $Q_8 = 80258/2809$         |
| $B_{9,2}$  | 835355/577368      | $Q_8 = 525692/25281$       |
| $B_{10,1}$ | 14506973/1336500   | $Q_9 = 4713989/404496$     |
| $B_{10,2}$ | 202207748/21484375 | $Q_9 = 648969/280900$      |
| $B_{11,1}$ | 507845/261954      | $Q_{10} = 317585/1617984$  |
| $B_{11,2}$ | 25143284/1546875   | $Q_{10} = 8811001/1123600$ |
| $B_{11,3}$ | 151707121/11000000 | $Q_{10} = 426449/3441025$  |

Every entry is positive. Rule  $B$ , and the preceding  $B_0$  argument for the first row, therefore pay all bridge states at the event frequency, not at a later upper checkpoint.

The four  $G$ -rows use the already displayed global calculation (16). For clarity, here are the exact references needed by the state map;  $S = (193/53)^2 + j(j+1) - t^2$  and  $R = (38/539)t^3 - C$ .

| row      | $j$ | $C$ | $t$     | $(S, R)$                                |
|----------|-----|-----|---------|-----------------------------------------|
| $G_8$    | 7   | 40  | 208/25  | (67049/1755625, 5083656/8421875)        |
| $G_9$    | 8   | 53  | 923/100 | (1901439/28090000, 656778873/269500000) |
| $G_{10}$ | 9   | 73  | 254/25  | (61431/1755625, 7911557/8421875)        |
| $G_{11}$ | 10  | 89  | 111/10  | (14211/280900, 1999489/269500).         |

(30)

Both components of every pair are positive.

**Proposition 2** (Closed post-event domination). *For every  $7 \leq m \leq 11$  and every  $(\rho, K) \in \mathfrak{B}_m^+$ ,*

$$\widehat{C}_z(K) < \mathcal{W}(\rho, K).$$

*At  $K = k_{11}(\rho)$ , the original strict proxy also satisfies  $C_z(K) < \mathcal{W}(\rho, K)$ . Consequently every fixed, moving, or coincident proxy event in  $k_6 \leq K \leq k_{11}$  is paid at its own wall.*

*Proof.* Fix a band. The upper checkpoint inventory (27) contains every possible event in that band; the algebraic splits in (9) cover the two higher radial tails at their equality faces. Choose  $H, h, \tau$  only after all modes at the event have been inserted into (25). The simultaneous restrictions (28) and the mutually exclusive conditions in the thirty-eight-row table put the resulting state in exactly one row. Formula (26) proves its displayed cap, even if the modal set has gaps.

An  $L$ -row is paid by its localization witness and its positive  $P$ . A  $B$ -row is paid on both closed sides of its split by its positive  $P, Q$ . A  $G$ -row is paid by (30); the initial row is paid by  $B_{7,0}$ . These alternatives exhaust the table, proving the closed post-event inequality.

No ordering premise has entered the argument. Changing the order of two fixed or moving walls merely changes which disjoint state is selected. At a branch equality in the maximum defining  $b_{\ell,n}$ , both lower walls remain valid. At a ratio face the defining state is impossible by the strict positive localization witness. At a bridge face both payment arguments hold with positive reserve.

At an interior modal wall the original strict proxy assigns equality to the pre-event cap, whereas the stronger closed post-event cap has just been paid. At a lower checkpoint  $K = k_{m-1}$ , the closed post-event version is assigned to the following half-open band and is covered even when  $K = k_{m-1}$ ; the original strict proxy omits every channel on its defining equality and remains owned by the preceding closed spectral band. At intermediate upper checkpoints the same convention passes the post-event state to the following band. Finally, at  $K = k_{11}$  the strict inequalities  $b_{\ell,n} < K$  give the same state restrictions and the  $m = 11$  payment rows directly, while the equality channel (11, 1) is absent. Coincident walls are harmless because (25) inserts their combined multiplicity before the unique state row is chosen.  $\square$

The proposition above is the final mathematical exhaustion theorem. It is not a row-count inference from the historical 611-predicate executable census. The map below is retained only as archival provenance for the older arithmetic implementation and is not a premise of Proposition 2.

## 10 Archival exact row map and scope audit

The following map uses the ordering of the two functions in the original exact registry. To keep the table readable,  $L$  denotes `verify_localizations`, and  $C$  denotes `verify_cap_tables_and_payments`. The map lets a referee match every replaced predicate to a displayed hand calculation.

| registry family | row numbers | analytic replacement here                                                                                                         |
|-----------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------|
| $L$             | 1–2         | distinctness and connectedness: (6)–(7) and the seventeen-row table                                                               |
| $L$             | 3–19        | all ratios in the high strip, by $\rho_c < 1/7 < 4/25$ and $2/3 < 7/8$                                                            |
| $L$             | 20–36       | the seventeen exact sign witnesses in the long table                                                                              |
| $L$             | 37–42       | split location (8) and side/equality arithmetic (9)                                                                               |
| $L$             | 43–47       | the five $k_{11}$ implications (10)–(13)                                                                                          |
| $C$             | 44–55       | the twelve non-bookkeeping cap–74 predicates, (20)–(23); row $C56$ , the block identity $49 + 25 = 74$ , is allocated to Packet A |
| $C$             | 57–59       | the three impossible cells, (14)                                                                                                  |
| $C$             | 78–79       | the general checkpoint monotonicity proof (15)                                                                                    |
| $C$             | 80–84       | the five global-payment rows following (16)                                                                                       |
| $C$             | 85–98       | the seven localization and seven conditional-payment rows in the final long table                                                 |

The count reconciliation is

| subfamily                     | connected mathematical objects | registry predicates |
|-------------------------------|--------------------------------|---------------------|
| registry metadata             | 1                              | 2                   |
| high-strip membership         | 17                             | 17                  |
| localization signs            | 17                             | 17                  |
| algebraic splits              | 3                              | 6                   |
| special $k_{11}$ implications | 5                              | 5                   |
| cap–74 path                   | 1                              | 12                  |
| impossible cells              | 3                              | 3                   |
| checkpoint monotonicity       | 1                              | 2                   |
| global payments               | 5                              | 5                   |
| conditional payments          | 7                              | 14                  |
| total                         | 60                             | 83.                 |

In particular, the five headline Packet–C subfamilies contain exactly

$$17 + 3 + 3 + 5 + 7 = 35$$

connected objects (localization signs, algebraic splits, impossible cells, global payments, and conditional payments). Because each algebraic split has a location and a side predicate, and each conditional payment has a localization and a reserve predicate, these headline objects replace  $17 + 6 + 3 + 5 + 14 = 45$  registry predicates. The remaining 38 predicates in the total are the explicitly listed auxiliary closure rows.

Thus this packet replaces all 47 predicates of  $L$  and 36 predicates of  $C$ . The elementary cap addition rows 1–43, row 56, and rows 60–77 of the latter family are outside the declared Packet C scope; they are instances of the general odd-sum/block-cap identity and are not claimed here. No row in the declared scope is unreconstructed.